



Prabhat Kumar

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EDUCATION

Indian Institute of Technology, Kharagpur

Bachelor of Science in Mathematics and Computing

CGPA: 7.29

EXPERIENCE

Software Engineering Intern

May 2026 – July 2026

Paycraft

github.com/Prabhat-190/pbf-generator

- Built a Java 21 Spring Boot service that generates ABT-to-NLP settlement files in Header, Record, and Trailer format. Validated transit transactions and tracked every file from creation through loyalty-callback completion.
- Designed REST APIs for generation, status lookup, callbacks, and health checks, then stored results with Spring Data JPA. Persisted file metadata, per-record outcomes, and four-digit daily sequences in an H2 database.
- Implemented trailer checks, SHA-256 checksums, ZIP packaging, mock SFTP delivery, and year-month archival. Added SLF4J request-id logging, OpenAPI docs, JUnit 5 tests, and Docker packaging for Railway deployment.

PROJECTS

Bachelor's Thesis Project: Machine Learning-Based Stock Prediction and Portfolio Optimization

Ongoing

Python, Pandas, NumPy, Scikit-learn, Keras, Matplotlib

- Building a machine learning pipeline that forecasts next-day Nifty 50 returns and then forms an optimized stock portfolio. Feeding historical OHLCV prices and technical indicators into the forecast stage before capital is allocated.
- Training Linear Regression, Random Forest, and LSTM models with walk-forward validation to block look-ahead leakage. Ranking mean-variance portfolios by Sharpe ratio and maximum drawdown against equal-weight and index baselines.

Algo Trade Simulator

Live Project

Python, Dash, Plotly, Redis, WebSockets, Docker, Gunicorn, Nginx

- Developed a live trading simulator that reads Level-2 order-book streams and estimates execution cost for market orders. Modeled slippage, market impact, exchange fees, liquidity, and volatility inside the cost estimate.
- Designed a four-layer Redis publish-subscribe pipeline for ingestion, book updates, cost analysis, and live charts. Deployed with Gunicorn and Nginx, added Redis failover, and wrote more than 100 tests for feeds and persistence.

Amazon FinAI: Smart Checkout and Conversational Assistant

GitHub

React, Node.js, FastAPI, MongoDB, Redis, ChromaDB, OpenAI, Docker

- Developed a checkout platform that routes payments and answers user questions with a retrieval-augmented assistant. Combined wallet, budget, and policy context so the assistant can return grounded financial answers.
- Ranked payment methods by gateway success, cashback, and gift-card expiry using cached Redis signals. Extracted discounts from bank-offer images with GPT-4 Vision and ran three services with Docker Compose.

Fraud Shield AI

Live Project

Python, Scikit-learn, Pandas, NumPy, Streamlit, CCXT

- Built a fraud detection model that scores each transaction from 0 to 100 percent risk using a Random Forest classifier. Tuned the decision threshold on the precision-recall curve so genuine payments are not blocked by false alerts.
- Engineered amount, time, cardholder age, location, and merchant-category features for each scored transaction. Deployed a Streamlit dashboard for live scoring and used CCXT market data with an automatic offline fallback.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Frontend: React.js, Next.js, HTML, CSS

Backend: Spring Boot, Spring Data JPA, REST APIs, Node.js, Express.js, FastAPI, WebSocket, Microservices

Machine Learning: Scikit-learn, Keras, Pandas, NumPy, LSTM, Time-Series Forecasting, Portfolio Optimization

Databases and Tools: MySQL, MongoDB, Redis, H2, ChromaDB, Docker, Git, Gradle, JUnit, SLF4J, Gunicorn, Nginx, OpenAPI, Postman

Computer Science: Object-Oriented Programming, Operating Systems, Computer Networks, System Design

RELEVANT COURSEWORK

Computer Science: Algorithms, Data Structures, DBMS, File Organization, Computer Organization, Theory of Computation, Big Data Analysis.

Mathematics: Optimization, Probability and Statistics, Stochastic Processes, Linear Algebra, Discrete Mathematics, Graph Theory.

ACHIEVEMENTS

- Rated Codeforces Expert with a maximum rating of 1736 and Global Rank 245 among 25,673 participants in Round 1093.
- Solved more than 500 data structures and algorithms problems on LeetCode, Codeforces, CodeChef, and HackerRank.
- Awarded the Reliance Foundation Undergraduate Scholarship, selected among the top 0.8 percent of over 120,000 applicants.
- Secured a Top 8 percent JEE Advanced rank, a Top 1.2 percent JEE Main rank, and the HackerRank Intern certification.